SYNCHRONOUS COUNTER

4-BINARY COUNTER

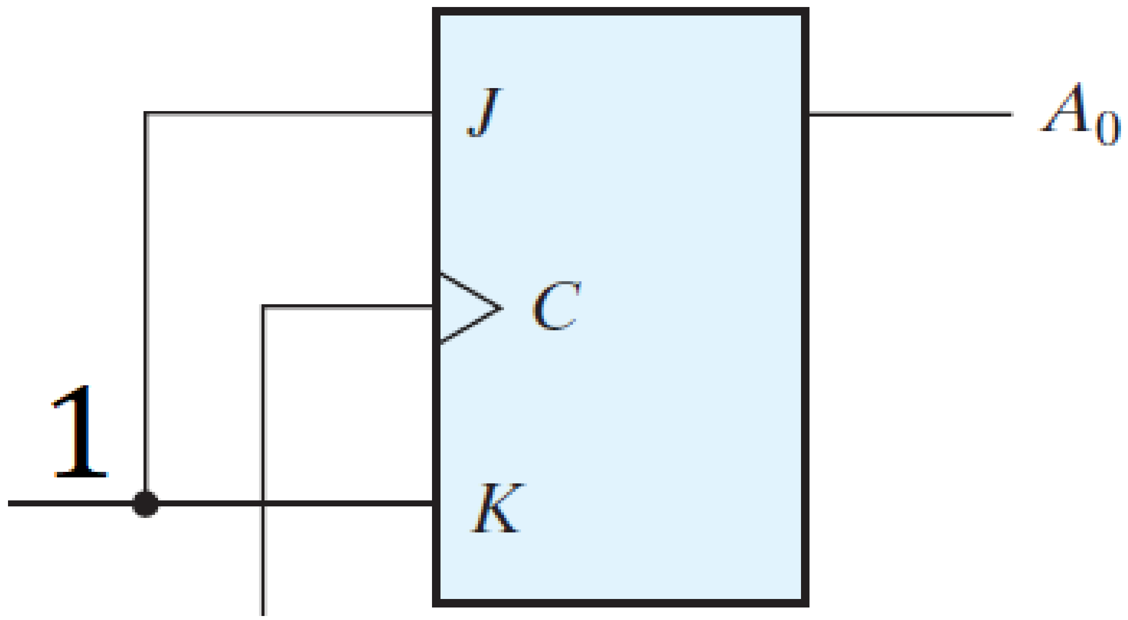
Lecture 11

Part 1

4-BIT UP BINARY COUNTER

CLK	A_3	A_2	A_1	A_0
1	0	0	0	0
2	0	0	0	1
3	0	0	1	0
4	0	0	1	1
5	0	1	0	0
6	0	1	0	1
7	0	1	1	0
8	0	1	1	1
9	1	0	0	0
10	1	0	0	1
11	1	0	1	0
12	1	0	1	1
13	1	1	0	0
14	1	1	0	1
15	1	1	1	0
16	1	1	1	1

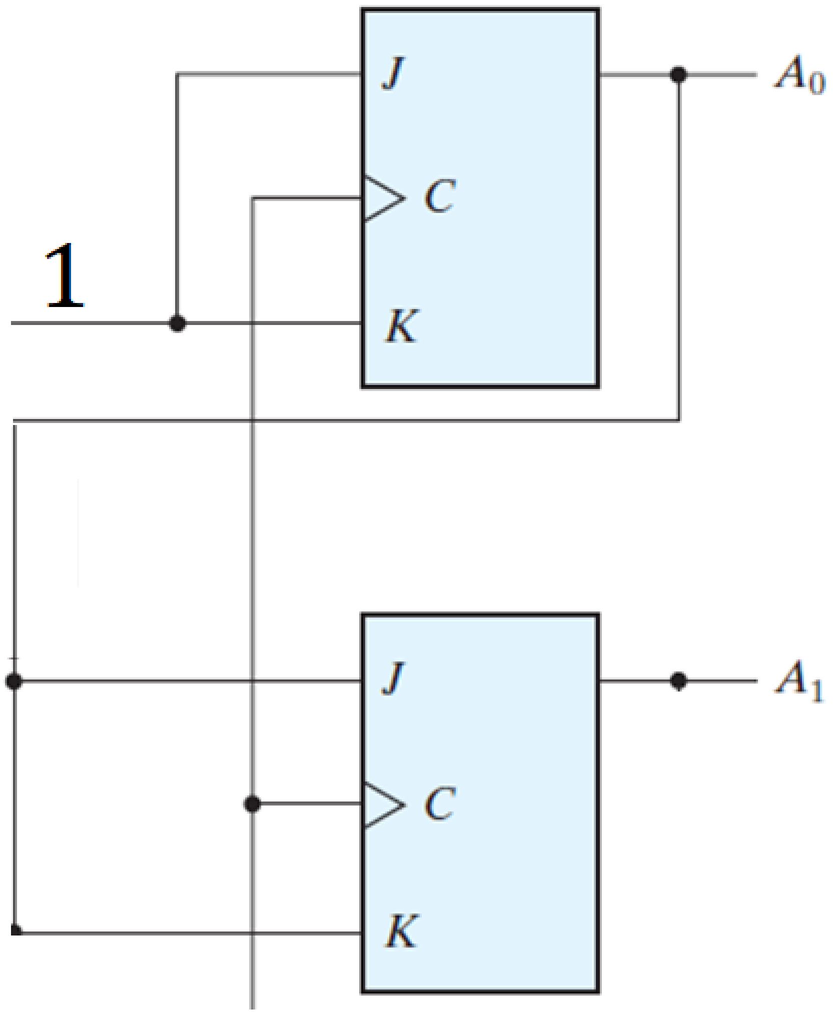
4-BIT UP BINARY COUNTER



4-BIT UP BINARY COUNTER

CLK	A_3	A_2	A_1	A_0
1	0	0	0	0
2	0	0	0	1
3	0	0	1	0
4	0	0	1	1
5	0	1	0	0
6	0	1	0	1
7	0	1	1	0
8	0	1	1	1
9	1	0	0	0
10	1	0	0	1
11	1	0	1	0
12	1	0	1	1
13	1	1	0	0
14	1	1	0	1
15	1	1	1	0
16	1	1	1	1

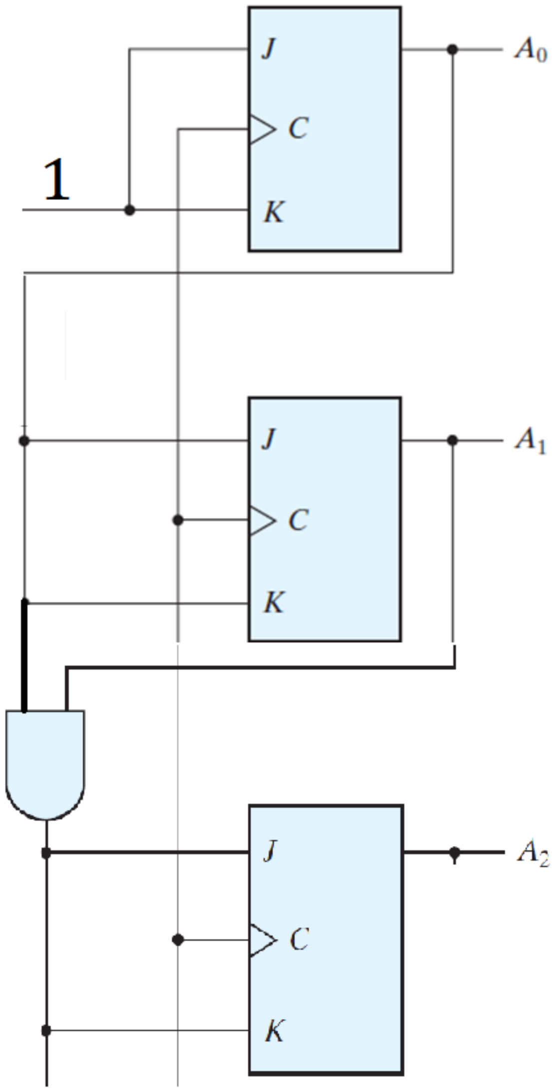
4-BIT UP BINARY COUNTER



4-BIT UP BINARY COUNTER

CLK	A_3	A_2	A_1	A_0
1,17	0	0	0	0
2	0	0	0	1
3	0	0	1	0
4	0	0	1	1
5	0	1	0	0
6	0	1	0	1
7	0	1	1	0
8	0	1	1	1
9	1	0	0	0
10	1	0	0	1
11	1	0	1	0
12	1	0	1	1
13	1	1	0	0
14	1	1	0	1
15	1	1	1	0
16	1	1	1	1

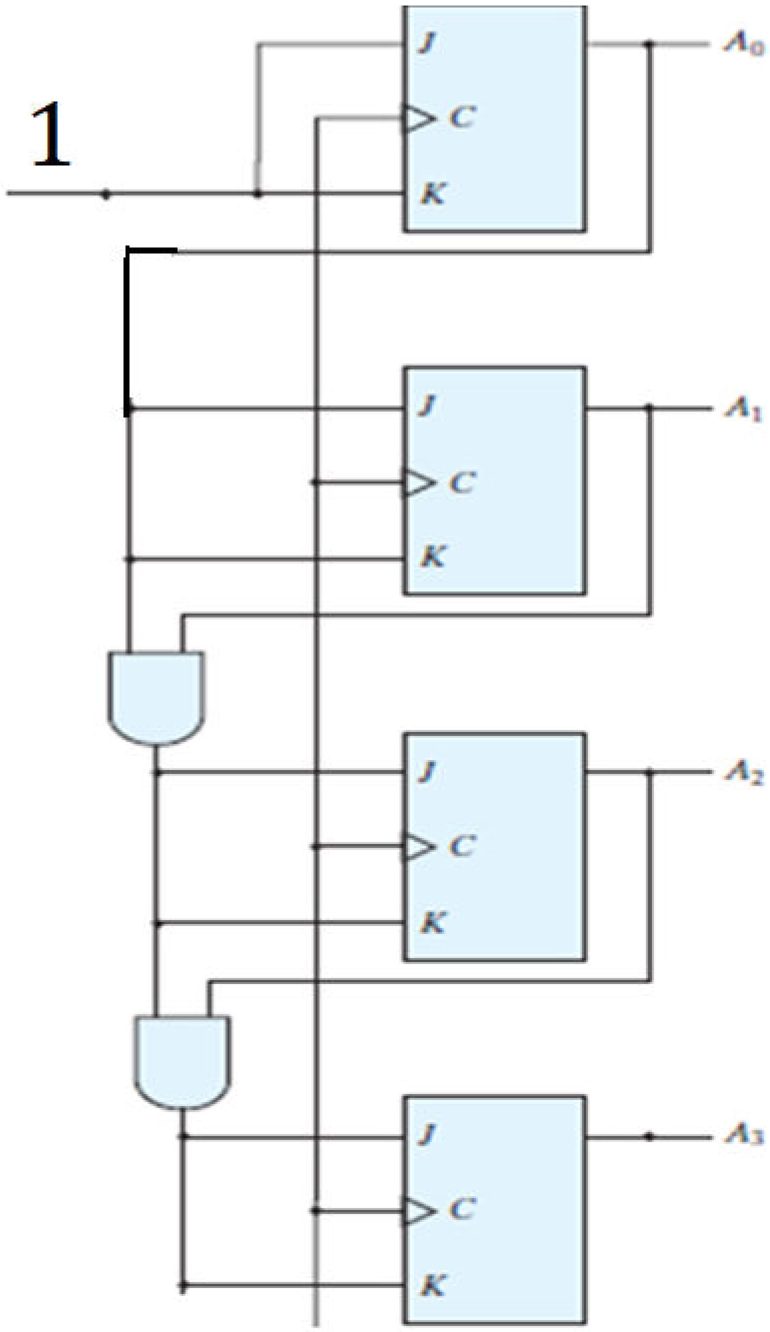
4-BIT UP BINARY COUNTER



4-BIT UP BINARY COUNTER

CLK	A_3	A_2	A_1	A_0
1,17	0	0	0	0
2	0	0	0	1
3	0	0	1	0
4	0	0	1	1
5	0	1	0	0
6	0	1	0	1
7	0	1	1	0
8	0	1	1	1
9	1	0	0	0
10	1	0	0	1
11	1	0	1	0
12	1	0	1	1
13	1	1	0	0
14	1	1	0	1
15	1	1	1	0
16	1	1	1	1

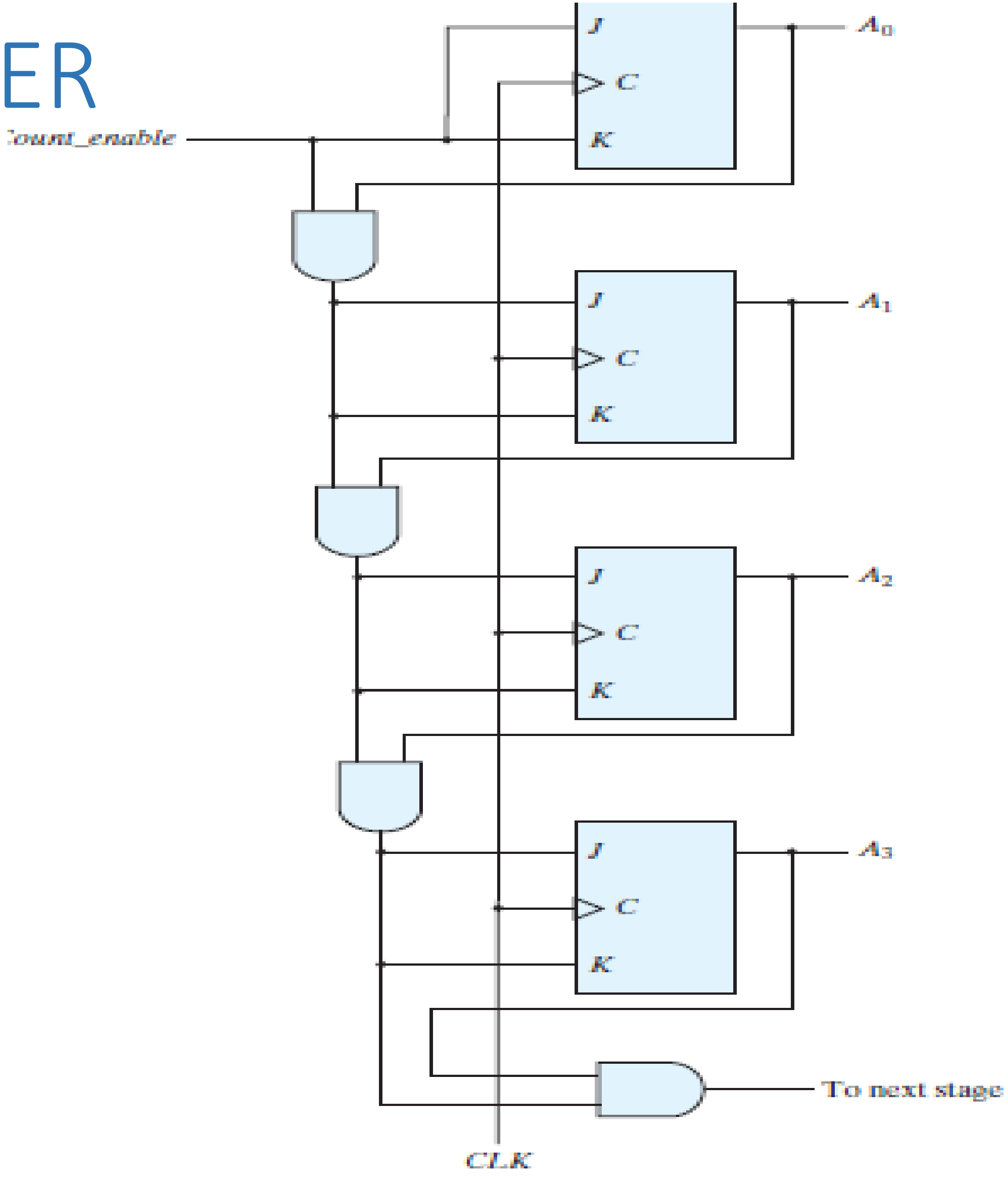
4-BIT UP BINARY COUNTER



4-BIT UP BINARY COUNTER

WITH

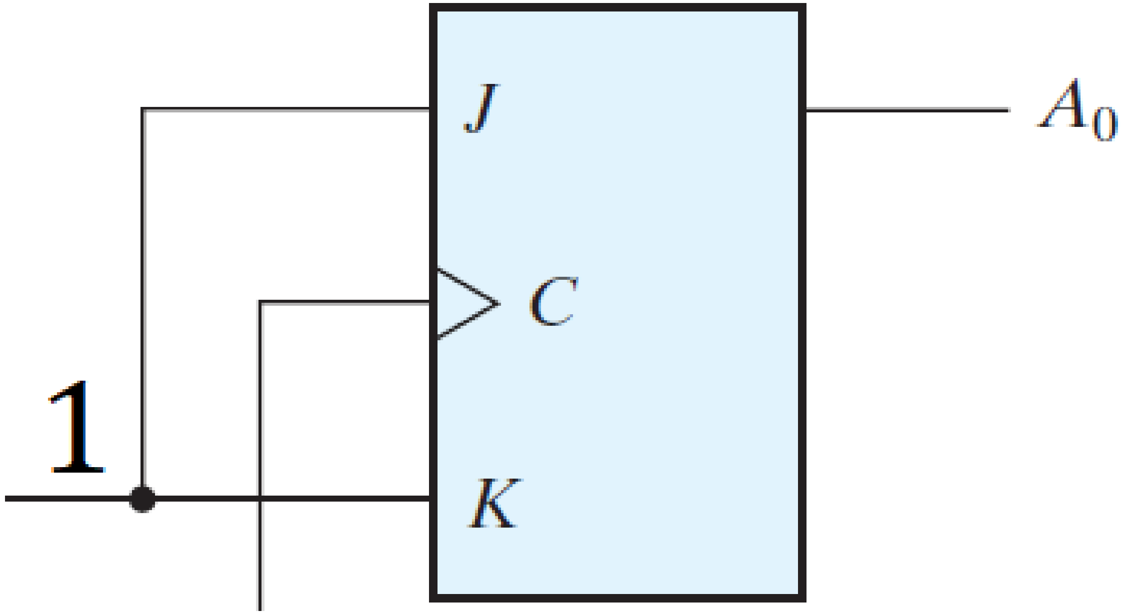
ENABLE



4-BIT DOWN BINARY COUNTER

CLK	A_3	A_2	A_1	A_0
1	1	1	1	1
2	1	1	1	0
3	1	1	0	1
4	1	1	0	0
5	1	0	1	1
6	1	0	1	0
7	1	0	0	1
8	1	0	0	0
9	0	1	1	1
10	0	1	1	0
11	0	1	0	1
12	0	1	0	0
13	0	0	1	1
14	0	0	1	0
15	0	0	0	1
16	0	0	0	0

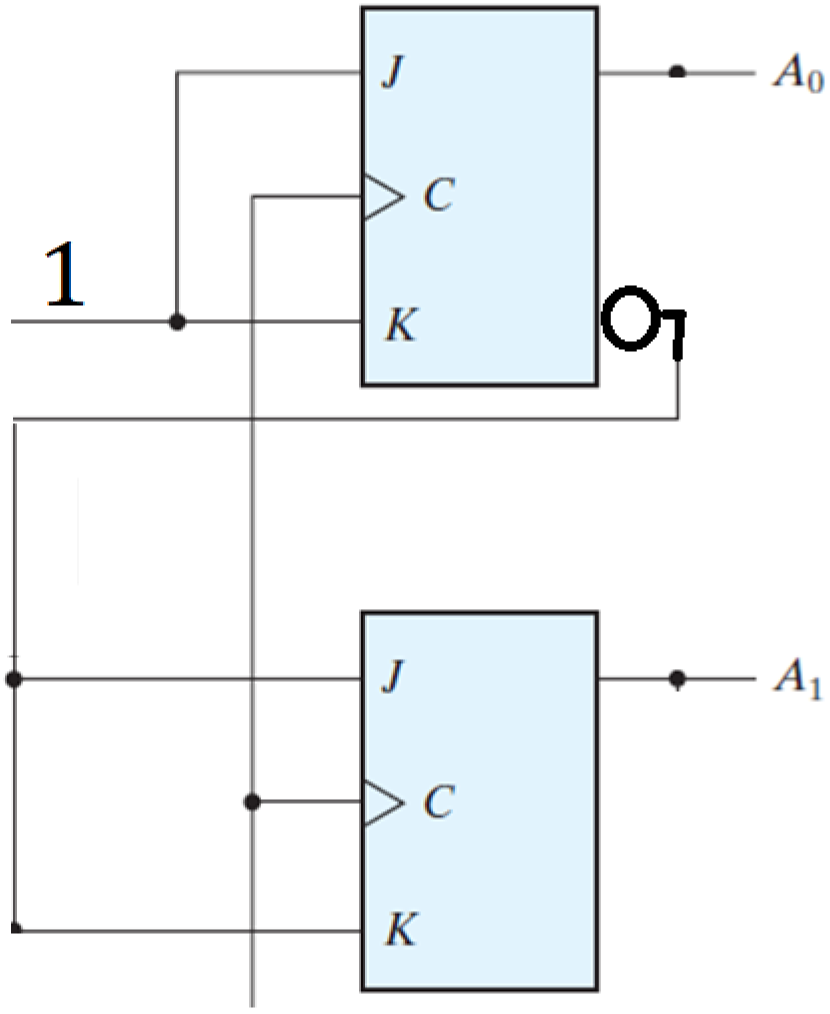
4-BIT DOWN BINARY COUNTER



4-BIT UP BINARY COUNTER

CLK	A_3	A_2	A_1	A_0
1	1	1	1	1
2	1	1	1	0
3	1	1	0	1
4	1	1	0	0
5	1	0	1	1
6	1	0	1	0
7	1	0	0	1
8	1	0	0	0
9	0	1	1	1
10	0	1	1	0
11	0	1	0	1
12	0	1	0	0
13	0	0	1	1
14	0	0	1	0
15	0	0	0	1
16	0	0	0	0

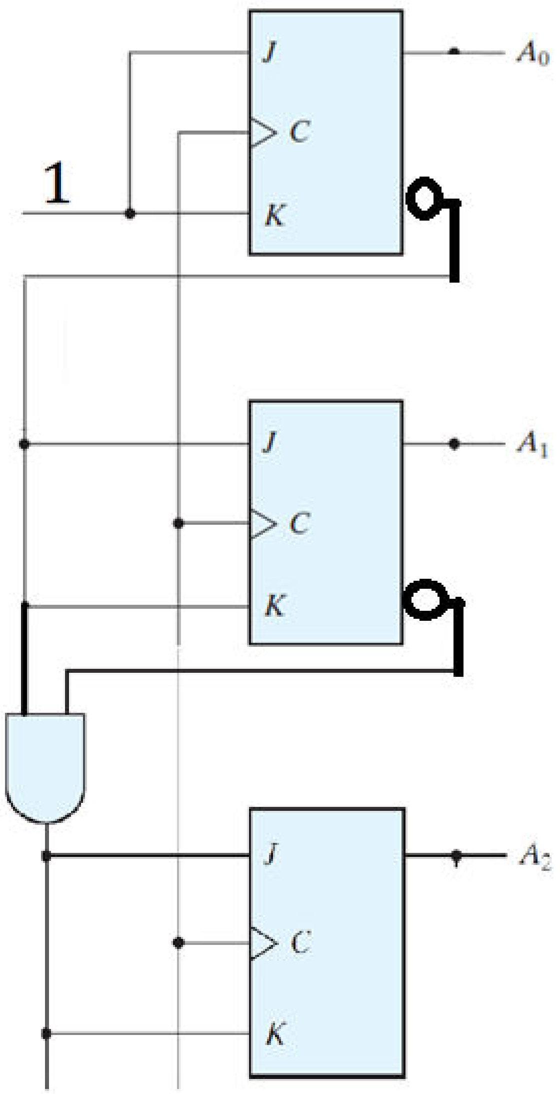
4-BIT BINARY COUNTER



4-BIT UP BINARY COUNTER

CLK	A_3	A_2	A_1	A_0
1,17	1	1	1	1
2	1	1	1	0
3	1	1	0	1
4	1	1	0	0
5	1	0	1	1
6	1	0	1	0
7	1	0	0	1
8	1	0	0	0
9	0	1	1	1
10	0	1	1	0
11	0	1	0	1
12	0	1	0	0
13	0	0	1	1
14	0	0	1	0
15	0	0	0	1
16	0	0	0	0

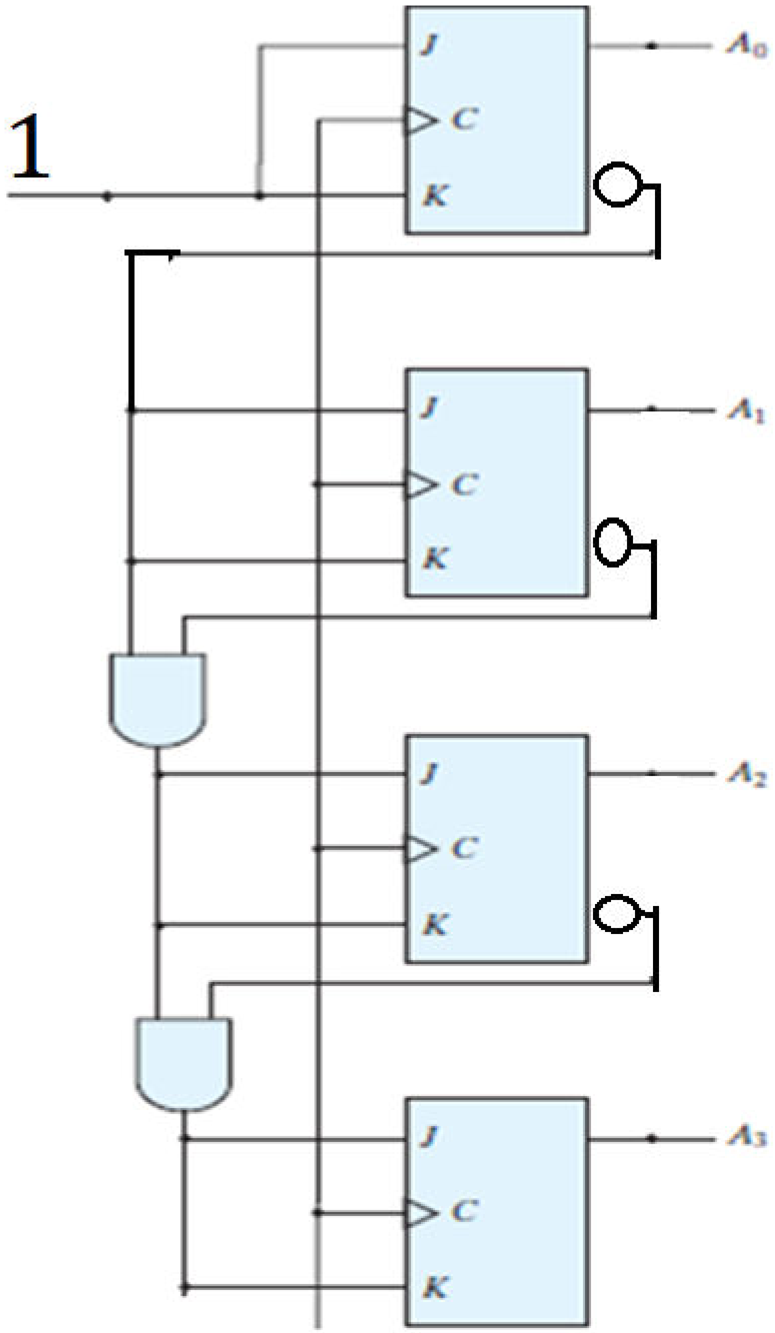
4-BIT BINARY COUNTER



4-BIT UP BINARY COUNTER

CLK	A_3	A_2	A_1	A_0
1,17	1	1	1	1
2	1	1	1	0
3	1	1	0	1
4	1	1	0	0
5	1	0	1	1
6	1	0	1	0
7	1	0	0	1
8	1	0	0	0
9	0	1	1	1
10	0	1	1	0
11	0	1	0	1
12	0	1	0	0
13	0	0	1	1
14	0	0	1	0
15	0	0	0	1
16	0	0	0	0

4-BIT BINARY COUNTER

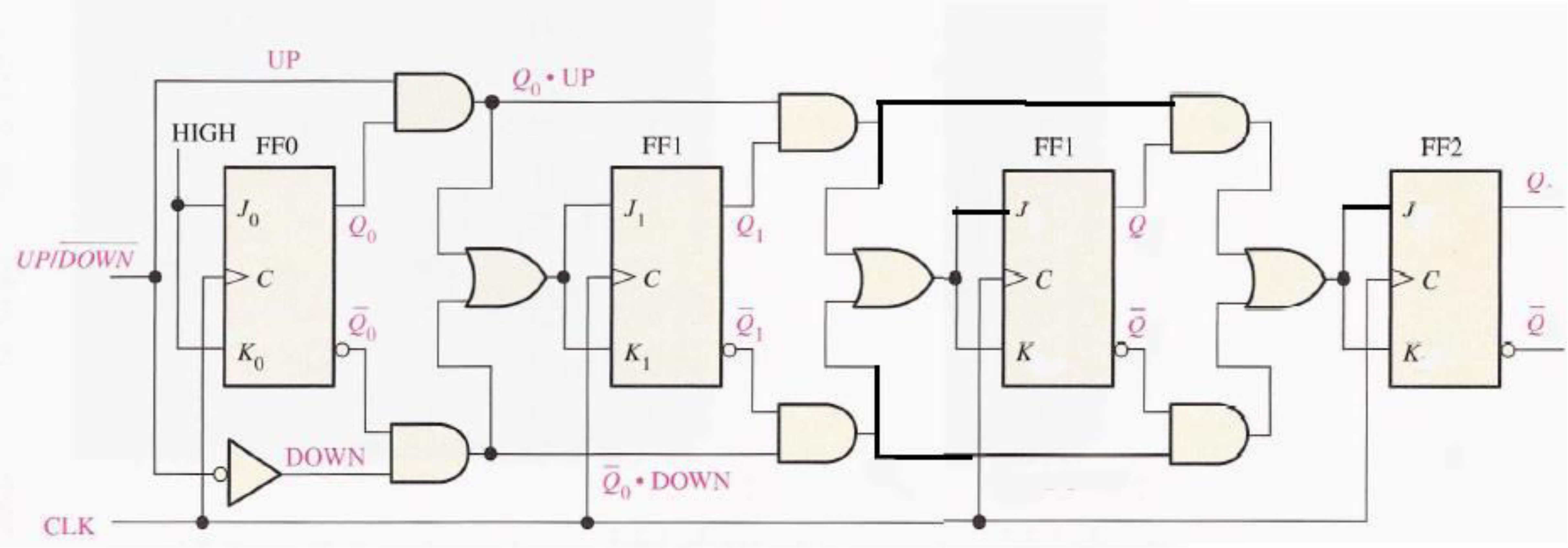


BCD SYNCHRONOUS COUNTER

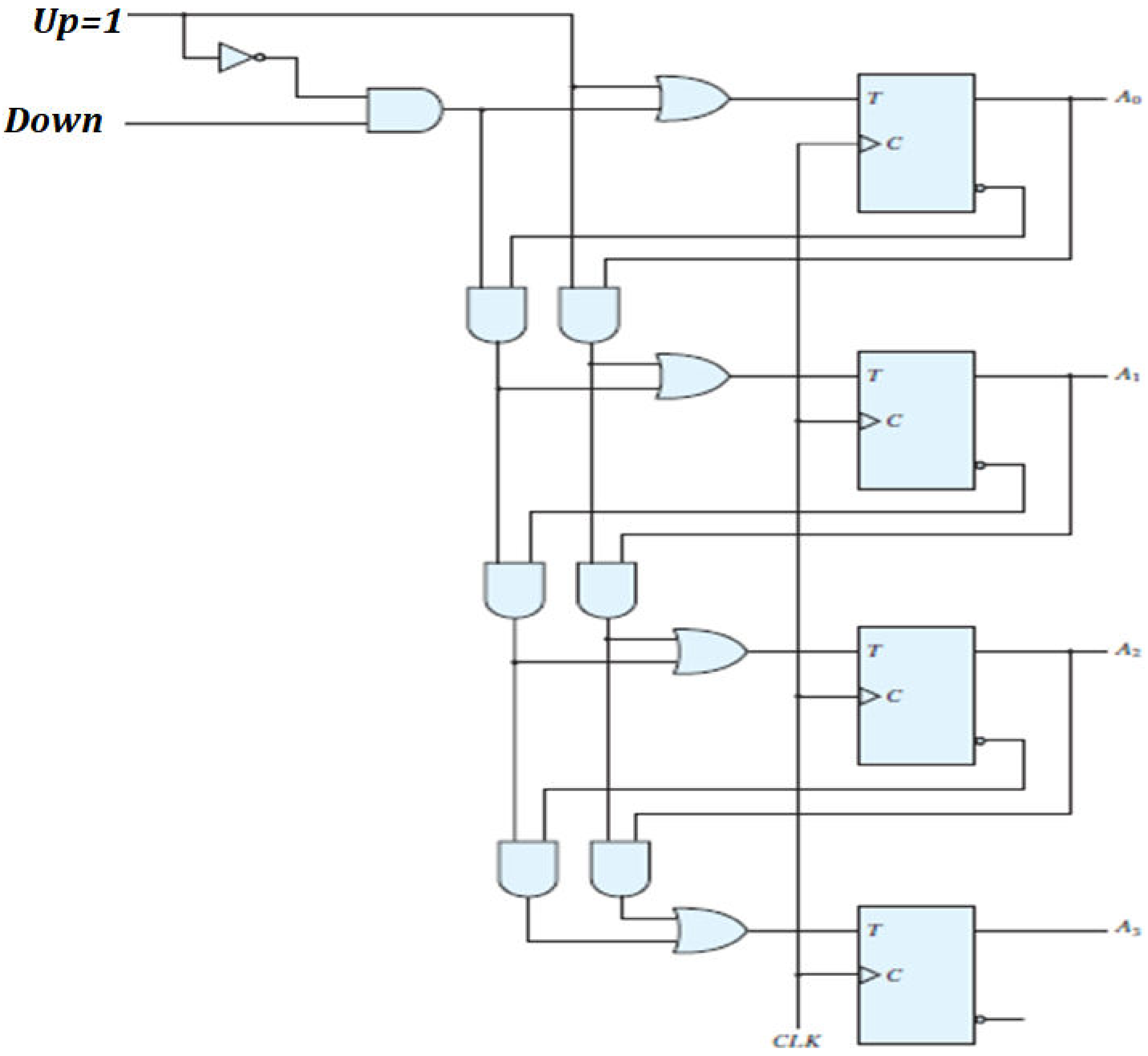
LECTURE 10

Part 2

4-BIT UP/DOWN COUNTER



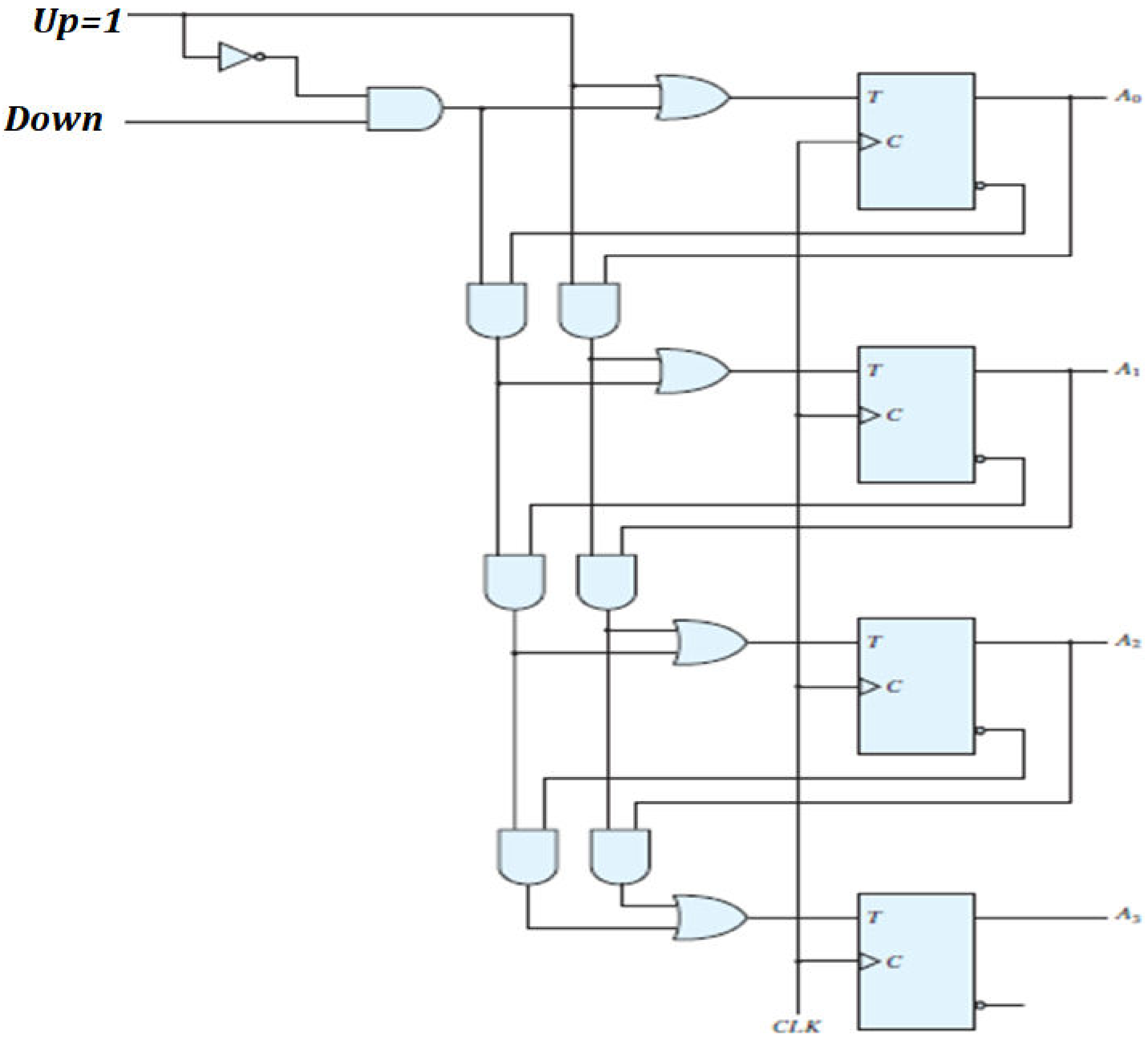
4-BIT UP/DOWN COUNTER



4-BIT UP/DOWN COUNTER

UP	DOWN	COMMENT
0	0	
0	1	
1	0	
1	1	

4-BIT UP/DOWN COUNTER



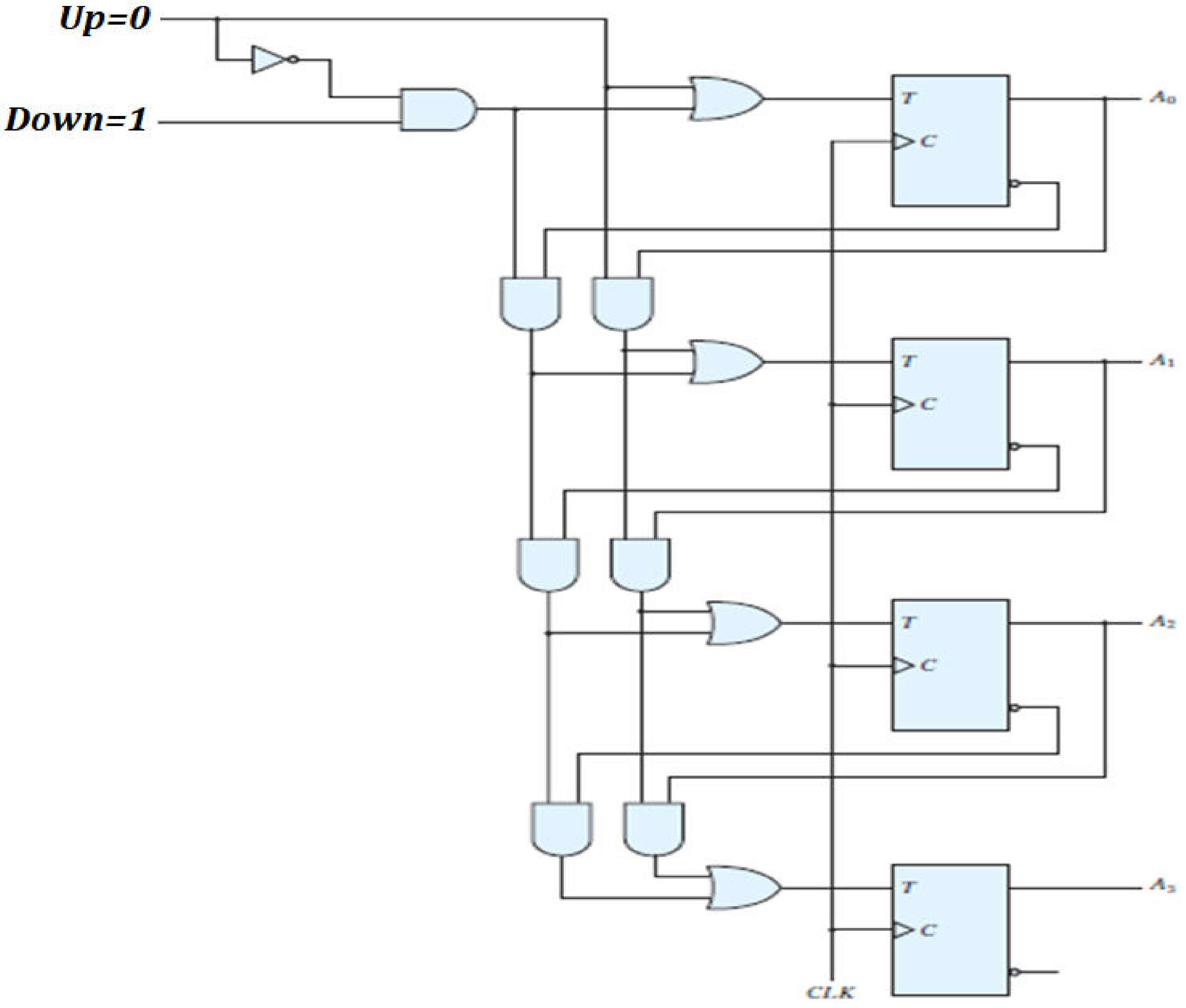
4-BIT UP/DOWN COUNTER

UP	DOWN	COMMENT
0	0	
0	1	
1	X	**UP**

4-BIT UP/DOWN COUNTER

UP	DOWN	COMMENT
0	0	
0	1	
1	X	**UP**

4-BIT UP/DOWN COUNTER



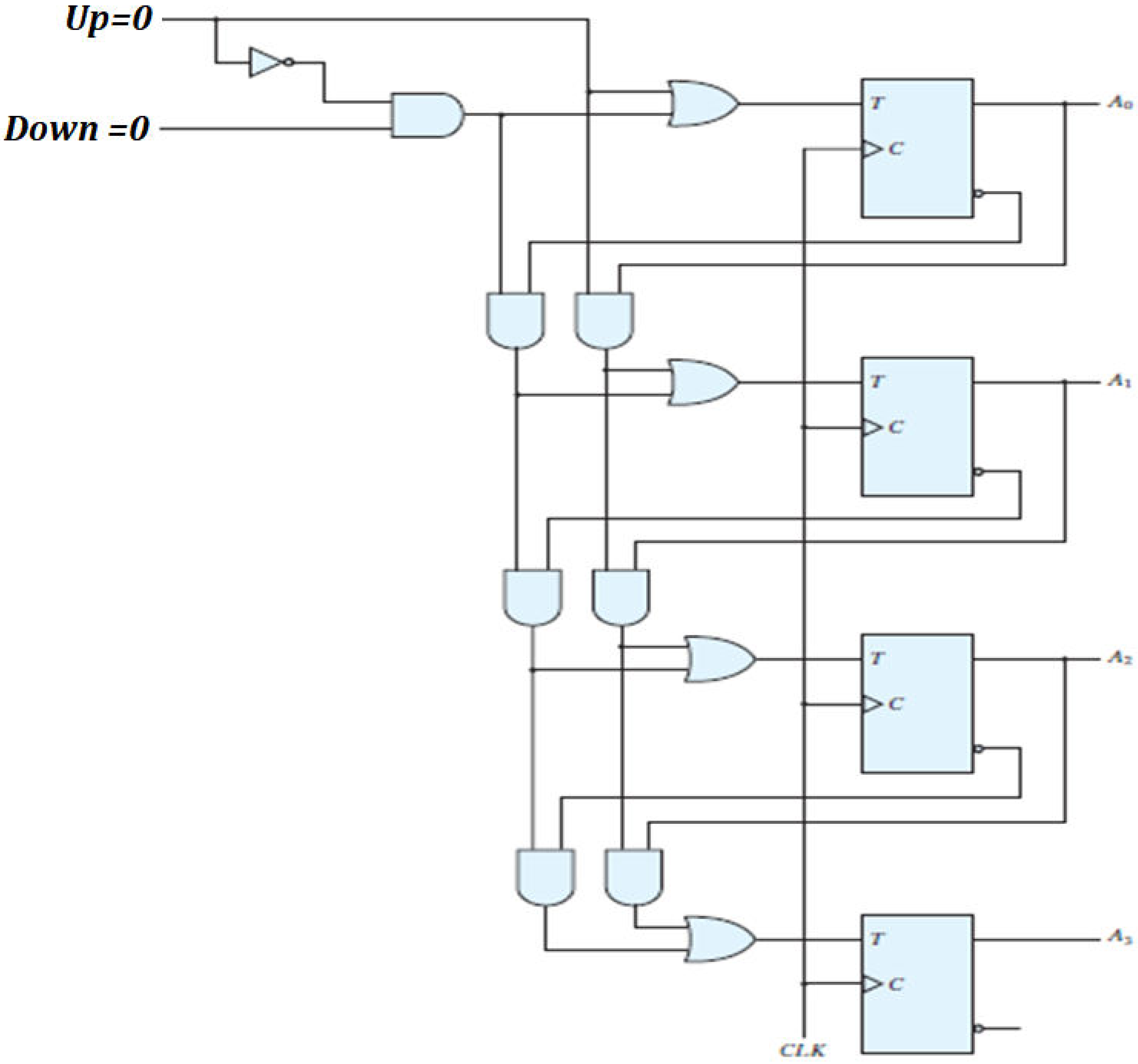
4-BIT UP/DOWN COUNTER

UP	DOWN	COMMENT
0	0	
0	1	**DOWN**
1	X	**UP**

4-BIT UP/DOWN COUNTER

UP	DOWN	COMMENT
0	0	
0	1	**DOWN**
1	X	**UP**

4-BIT UP/DOWN COUNTER



4-BIT UP/DOWN COUNTER

UP	DOWN	COMMENT
0	0	**NO CHANGE**
0	1	**DOWN**
1	X	**UP**

THANKS

BCD COUNTER

PRESENT STATE				NEXT STATE				OUTPUT
Q_8	Q_4	Q_2	Q_1	Q_8	Q_4	Q_2	Q_1	y
0	0	0	0	0	0	0	1	0
0	0	0	1	0	0	1	0	0
0	0	1	0	0	0	1	1	0
0	0	1	1	0	1	0	0	0
0	1	0	0	0	1	0	1	0
0	1	0	1	0	1	1	0	0
0	1	1	0	0	1	1	1	0
0	1	1	1	1	0	0	0	0
1	0	0	0	1	0	0	1	0
1	0	0	1	0	0	0	0	1

BCD COUNTER

PRESENT STATE				NEXT STATE				OUTPUT	FLIP FLOP INPUTS			
Q_8	Q_4	Q_2	Q_1	Q_8	Q_4	Q_2	Q_1	y	TQ_8	TQ_4	TQ_2	TQ_1
0	0	0	0	0	0	0	1	0				
0	0	0	1	0	0	1	0	0				
0	0	1	0	0	0	1	1	0				
0	0	1	1	0	1	0	0	0				
0	1	0	0	0	1	0	1	0				
0	1	0	1	0	1	1	0	0				
0	1	1	0	0	1	1	1	0				
0	1	1	1	1	0	0	0	0				
1	0	0	0	1	0	0	1	0				
1	0	0	1	0	0	0	0	1				

BCD COUNTER

PRESENT STATE				NEXT STATE				OUTPUT	FLIP FLOP INPUTS			
Q_8	Q_4	Q_2	Q_1	Q_8	Q_4	Q_2	Q_1	y	TQ_8	TQ_4	TQ_2	TQ_1
0	0	0	0	0	0	0	1	0				1
0	0	0	1	0	0	1	0	0			1	1
0	0	1	0	0	0	1	1	0				1
0	0	1	1	0	1	0	0	0		1	1	1
0	1	0	0	0	1	0	1	0				1
0	1	0	1	0	1	1	0	0			1	1
0	1	1	0	0	1	1	1	0				1
0	1	1	1	1	0	0	0	0	1	1	1	1
1	0	0	0	1	0	0	1	0				1
1	0	0	1	0	0	0	0	1	1			1

BCD COUNTER

PRESENT STATE				NEXT STATE				OUTPUT	FLIP FLOP INPUTS			
Q_8	Q_4	Q_2	Q_1	Q_8	Q_4	Q_2	Q_1	y	TQ_8	TQ_4	TQ_2	TQ_1
0	0	0	0	0	0	0	1	0	0	0	0	1
0	0	0	1	0	0	1	0	0	0	0	1	1
0	0	1	0	0	0	1	1	0	0	0	0	1
0	0	1	1	0	1	0	0	0	0	1	1	1
0	1	0	0	0	1	0	1	0	0	0	0	1
0	1	0	1	0	1	1	0	0	0	0	1	1
0	1	1	0	0	1	1	1	0	0	0	0	1
0	1	1	1	1	0	0	0	0	1	1	1	1
1	0	0	0	1	0	0	1	0	0	0	0	1
1	0	0	1	0	0	0	0	1	1	0	0	1

BCD COUNTER

PRESENT STATE				OUT PUT	FLIP FLOP INPUTS			
Q_8	Q_4	Q_2	Q_1	y	TQ_8	TQ_4	TQ_2	TQ_1
0	0	0	0	0	0	0	0	1
0	0	0	1	0	0	0	1	1
0	0	1	0	0	0	0	0	1
0	0	1	1	0	0	1	1	1
0	1	0	0	0	0	0	0	1
0	1	0	1	0	0	0	1	1
0	1	1	0	0	0	0	0	1
0	1	1	1	0	1	1	1	1
1	0	0	0	0	0	0	0	1
1	0	0	1	1	1	0	0	1

BCD COUNTER

TQ_1

	$Q_8Q_4\backslash Q_2Q_1$	$\overline{Q_2}\,\overline{Q_1}$ 00	$\overline{Q_2}Q_1$ 01	Q_2Q_1 11	$Q_2\overline{Q_1}$ 10
$\overline{Q_8}\,\overline{Q_4}$	00	1	1	1	1
$\overline{Q_8}Q_4$	01	1	1	1	1
Q_8Q_4	11	X	X	X	X
$Q_8\overline{Q_4}$	10	1	1	X	X

PRESENT STATE				OUT PUT	FLIP FLOP INPUTS			
Q_8	Q_4	Q_2	Q_1	y	TQ_8	TQ_4	TQ_2	TQ_1
0	0	0	0	0	0	0	0	1
0	0	0	1	0	0	0	1	1
0	0	1	0	0	0	0	0	1
0	0	1	1	0	0	1	1	1
0	1	0	0	0	0	0	0	1
0	1	0	1	0	0	0	1	1
0	1	1	0	0	0	0	0	1
0	1	1	1	0	1	1	1	1
1	0	0	0	0	0	0	0	1
1	0	0	1	1	1	0	0	1

$$TQ_1 = 1$$

BCD COUNTER

TQ_2		$\overline{Q_2}\,\overline{Q_1}$	$\overline{Q_2}Q_1$	Q_2Q_1	$Q_2\overline{Q_1}$
	$Q_8Q_4\backslash Q_2Q_1$	**00**	**01**	**11**	**10**
$\overline{Q_8}\,\overline{Q_4}$	**00**	0	1	1	0
$\overline{Q_8}Q_4$	**01**	0	1	1	0
Q_8Q_4	**11**	X	X	X	X
$Q_8\overline{Q_4}$	**10**	0	0	X	X

PRESENT STATE				OUT PUT	FLIP FLOP INPUTS			
Q_8	Q_4	Q_2	Q_1	y	TQ_8	TQ_4	TQ_2	TQ_1
0	0	0	0	0	0	0	0	1
0	0	0	1	0	0	0	1	1
0	0	1	0	0	0	0	0	1
0	0	1	1	0	0	1	1	1
0	1	0	0	0	0	0	0	1
0	1	0	1	0	0	0	1	1
0	1	1	0	0	0	0	0	1
0	1	1	1	0	1	1	1	1
1	0	0	0	0	0	0	0	1
1	0	0	1	1	1	0	0	1

$$TQ_2=\overline{Q_8}Q_1$$

BCD COUNTER

PRESENT STATE				OUT PUT	FLIP FLOP INPUTS			
Q_8	Q_4	Q_2	Q_1	y	TQ_8	TQ_4	TQ_2	TQ_1
0	0	0	0	0	0	0	0	1
0	0	0	1	0	0	0	1	1
0	0	1	0	0	0	0	0	1
0	0	1	1	0	0	1	1	1
0	1	0	0	0	0	0	0	1
0	1	0	1	0	0	0	1	1
0	1	1	0	0	0	0	0	1
0	1	1	1	0	1	1	1	1
1	0	0	0	0	0	0	0	1
1	0	0	1	1	1	0	0	1

TQ_4

		$\overline{Q_2}\,\overline{Q_1}$	$\overline{Q_2}Q_1$	Q_2Q_1	$Q_2\overline{Q_1}$
	Q_8Q_4\ Q_2Q_1	00	01	11	10
$\overline{Q_8}\,\overline{Q_4}$	00	0	0	1	0
$\overline{Q_8}Q_4$	01	0	0	1	0
Q_8Q_4	11	X	X	X	X
$Q_8\overline{Q_4}$	10	0	0	X	X

$$TQ_4 = Q_2Q_1$$

BCD COUNTER

TQ_8

		$\overline{Q_2}\,\overline{Q_1}$	$\overline{Q_2}Q_1$	Q_2Q_1	$Q_2\overline{Q_1}$
	$Q_8Q_4\backslash Q_2Q_1$	**00**	**01**	**11**	**10**
$\overline{Q_8}\,\overline{Q_4}$	**00**	0	0	0	0
$\overline{Q_8}Q_4$	**01**	0	0	1	0
Q_8Q_4	**11**	X	X	X	X
$Q_8\overline{Q_4}$	**10**	0	1	X	X

PRESENT STATE				OUT PUT	FLIP FLOP INPUTS			
Q_8	Q_4	Q_2	Q_1	y	TQ_8	TQ_4	TQ_2	TQ_1
0	0	0	0	0	0	0	0	1
0	0	0	1	0	0	0	1	1
0	0	1	0	0	0	0	0	1
0	0	1	1	0	0	1	1	1
0	1	0	0	0	0	0	0	1
0	1	0	1	0	0	0	1	1
0	1	1	0	0	0	0	0	1
0	1	1	1	0	1	1	1	1
1	0	0	0	0	0	0	0	1
1	0	0	1	1	1	0	0	1

$$TQ_8 = Q_4Q_2Q_1 + Q_2Q_1$$

BCD COUNTER

Present State				Next State				Output	Flip-Flop Inputs			
Q_8	Q_4	Q_2	Q_1	Q_8	Q_4	Q_2	Q_1	y	TQ_8	TQ_4	TQ_2	TQ_1
0	0	0	0	0	0	0	1	0	0	0	0	1
0	0	0	1	0	0	1	0	0	0	0	1	1
0	0	1	0	0	0	1	1	0	0	0	0	1
0	0	1	1	0	1	0	0	0	0	1	1	1
0	1	0	0	0	1	0	1	0	0	0	0	1
0	1	0	1	0	1	1	0	0	0	0	1	1
0	1	1	0	0	1	1	1	0	0	0	0	1
0	1	1	1	1	0	0	0	0	1	1	1	1
1	0	0	0	1	0	0	1	0	0	0	0	1
1	0	0	1	0	0	0	0	1	1	0	0	1

$$T_{Q1} = 1$$

$$T_{Q2} = Q'_8 Q_1$$

$$T_{Q4} = Q_2 Q_1$$

$$T_{Q8} = Q_8 Q_1 + Q_4 Q_2 Q_1$$

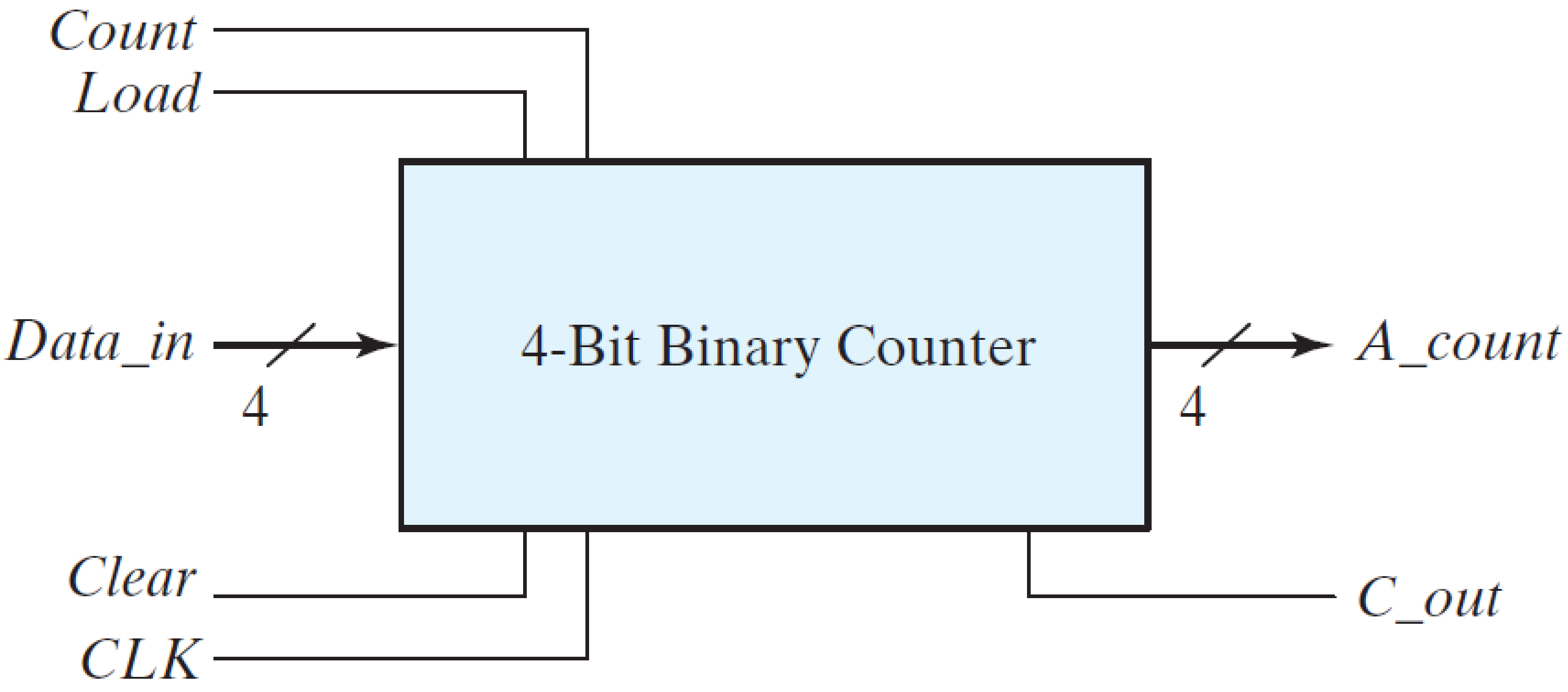
$$y = Q_8 Q_1$$

Synchronous Counter with PARALLEL LOAD

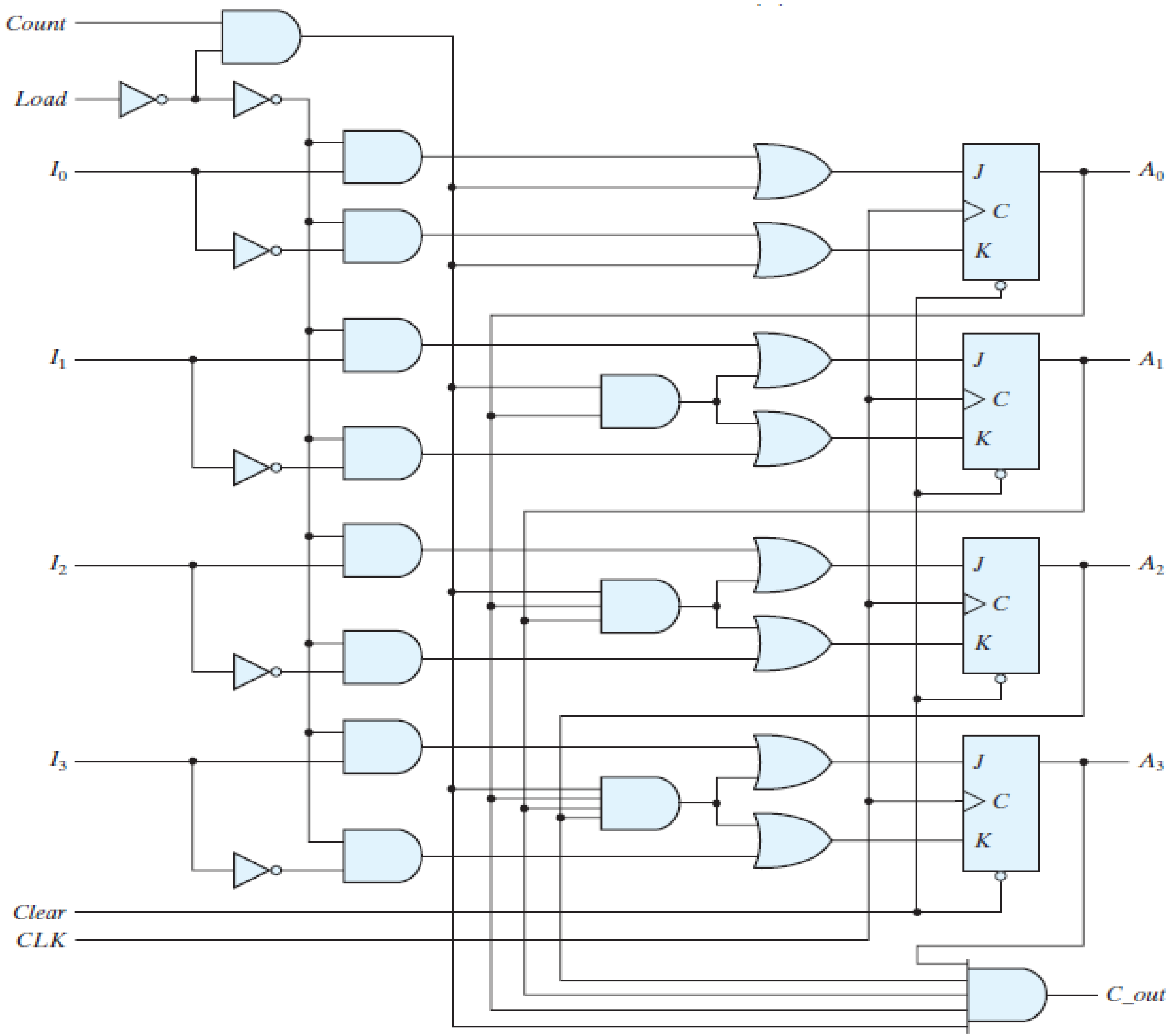
LECTURE 10

Part 3

BINARY COUNTER with PARALLEL LOAD



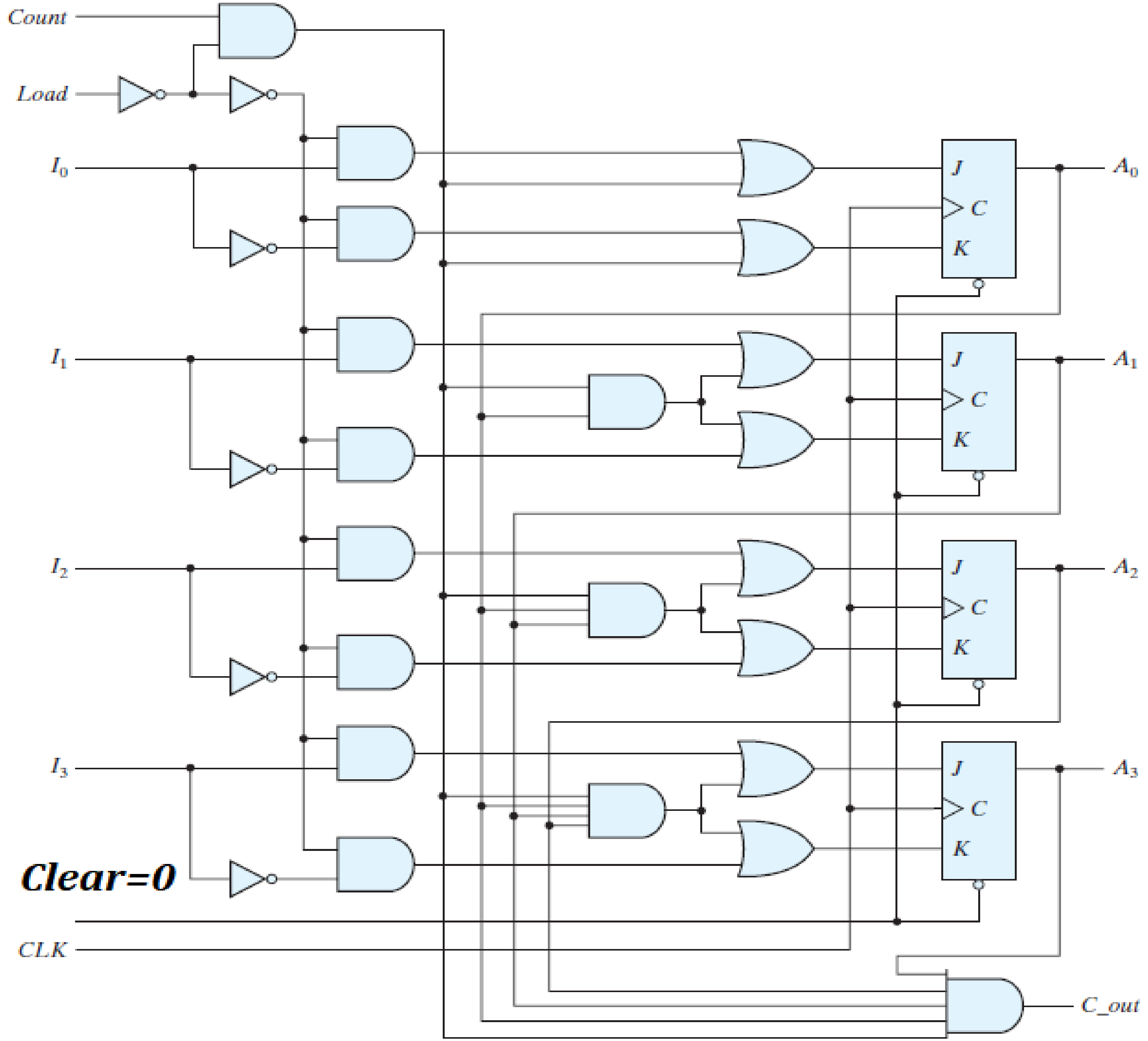
BINARY COUNTER with PARALLEL LOAD



BINARY COUNTER with PARALLEL LOAD

Clear	CLK	Load	Count	Function
0	X	X	X	
1	↑	1	X	
1	↑	0	1	
1	↑	0	0	

BINARY COUNTER with PARALLEL LOAD



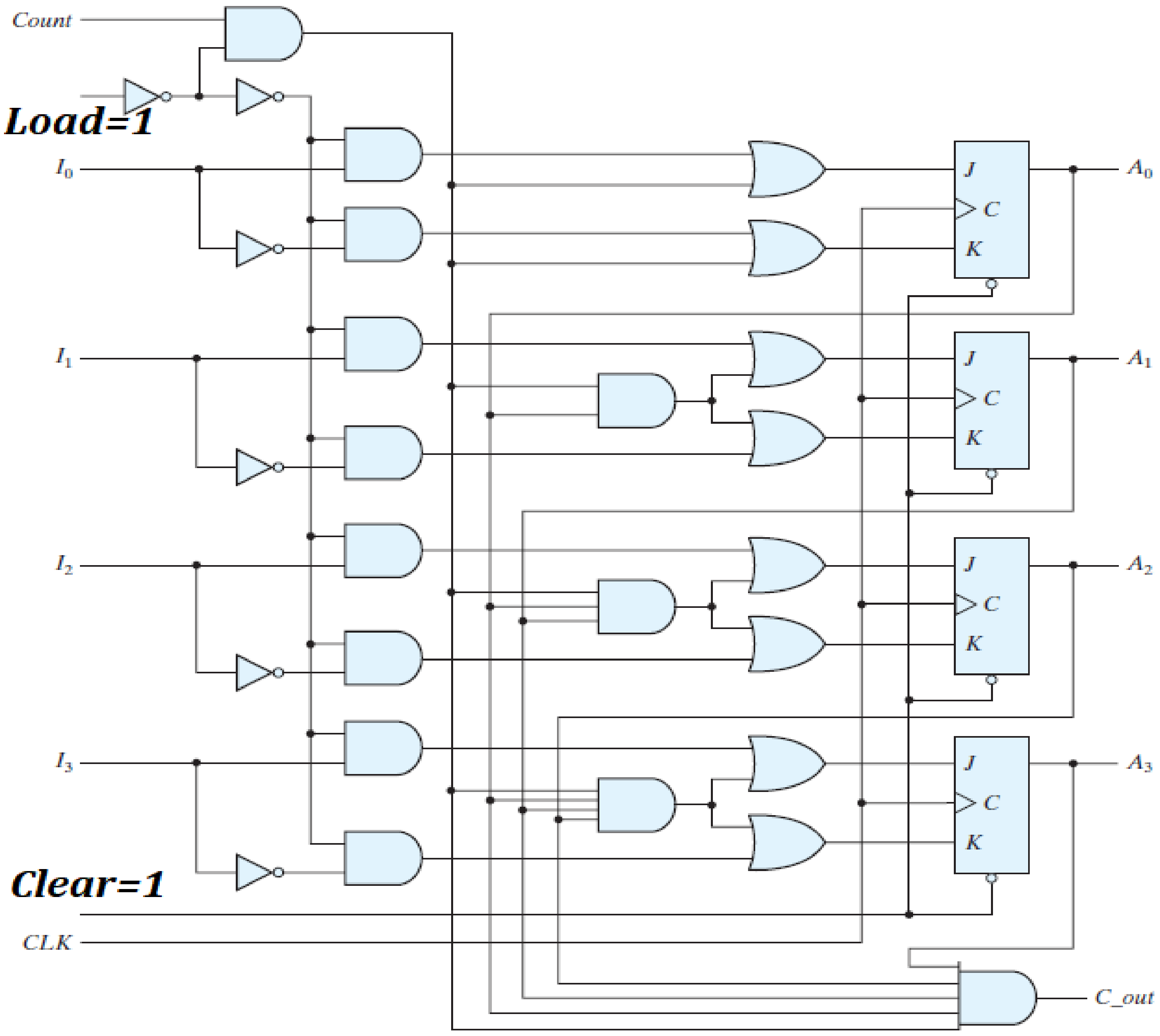
BINARY COUNTER with PARALLEL LOAD

Clear	CLK	Load	Count	Function
0	X	X	X	**CLEAR** to 0
1	↑	1	X	
1	↑	0	1	
1	↑	0	0	

BINARY COUNTER with PARALLEL LOAD

Clear	CLK	Load	Count	Function
0	X	X	X	**CLEAR** to 0
1	↑	1	X	
1	↑	0	1	
1	↑	0	0	

BINARY COUNTER with PARALLEL LOAD



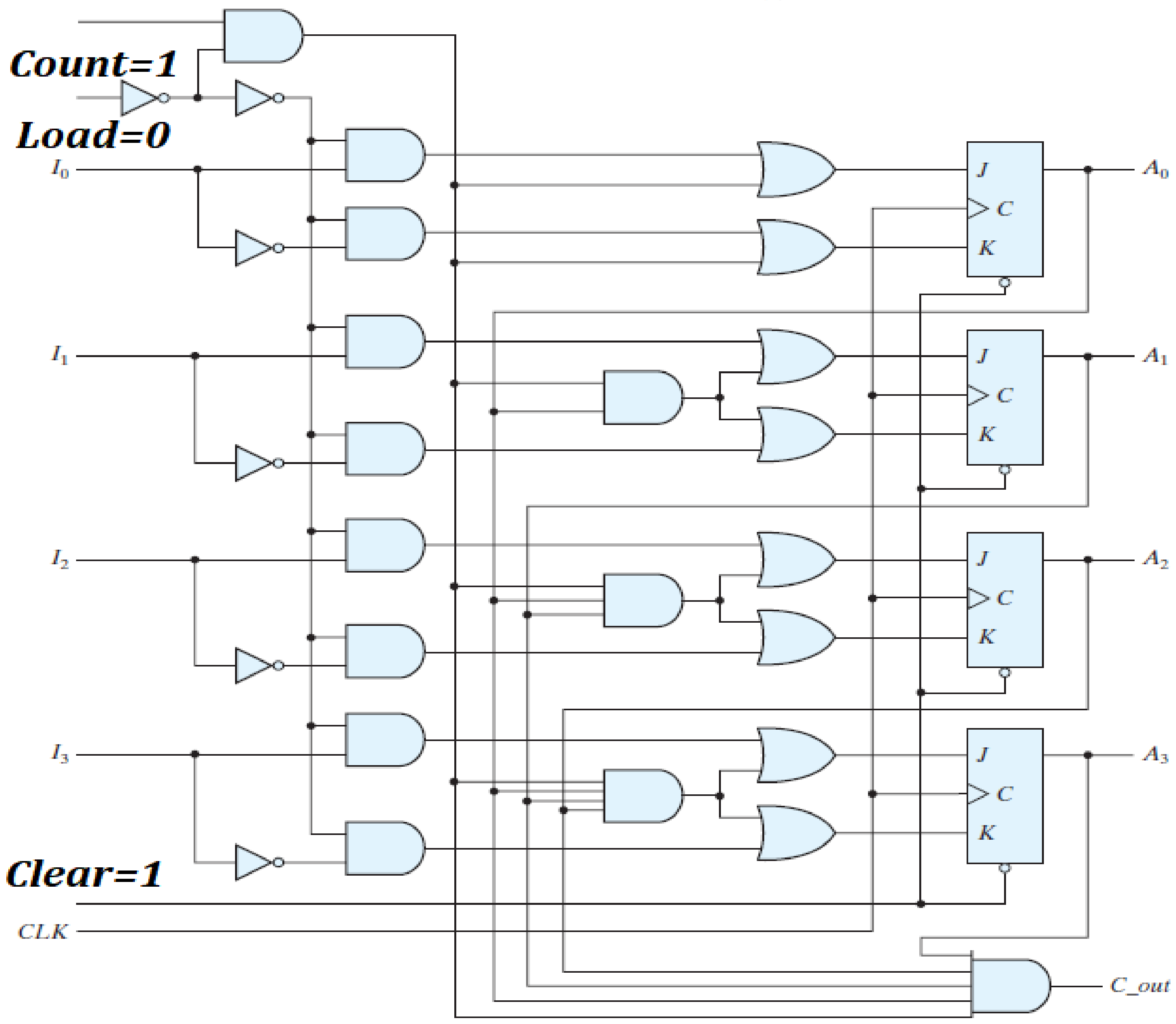
BINARY COUNTER with PARALLEL LOAD

Clear	CLK	Load	Count	Function
0	X	X	X	**CLEAR** to 0
1	↑	1	X	**LOAD** inputs
1	↑	0	1	
1	↑	0	0	

BINARY COUNTER with PARALLEL LOAD

Clear	CLK	Load	Count	Function
0	X	X	X	**CLEAR** to 0
1	↑	1	X	**LOAD** inputs
1	↑	0	1	
1	↑	0	0	

BINARY COUNTER with PARALLEL LOAD



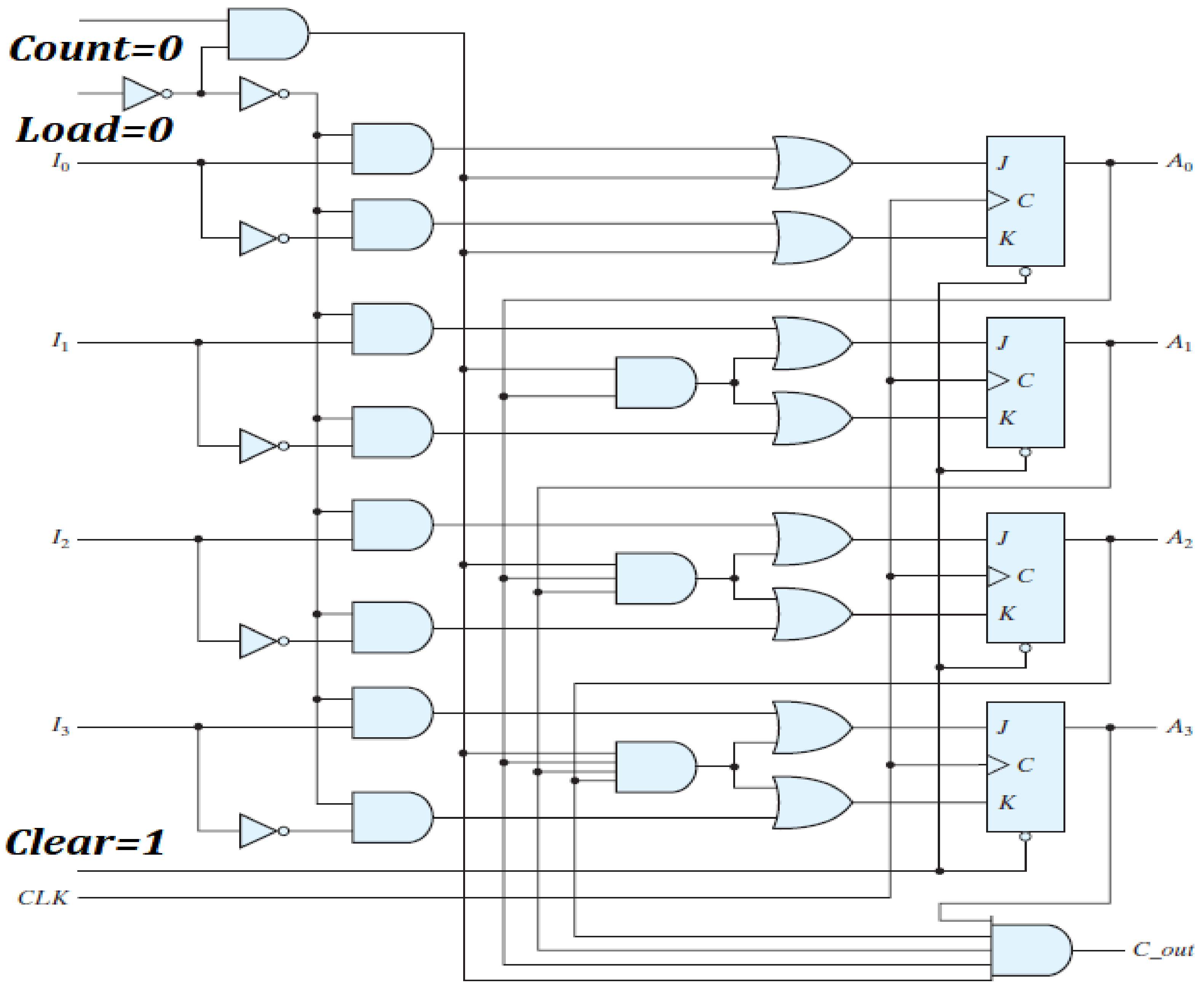
BINARY COUNTER with PARALLEL LOAD

Clear	CLK	Load	Count	Function
0	X	X	X	**CLEAR** to 0
1	↑	1	X	**LOAD** inputs
1	↑	0	1	**COUNT** to Next Binary state
1	↑	0	0	

BINARY COUNTER with PARALLEL LOAD

Clear	CLK	Load	Count	Function
0	X	X	X	**CLEAR** to 0
1	↑	1	X	**LOAD** inputs
1	↑	0	1	**COUNT** to Next Binary state
1	↑	0	0	

BINARY COUNTER with PARALLEL LOAD

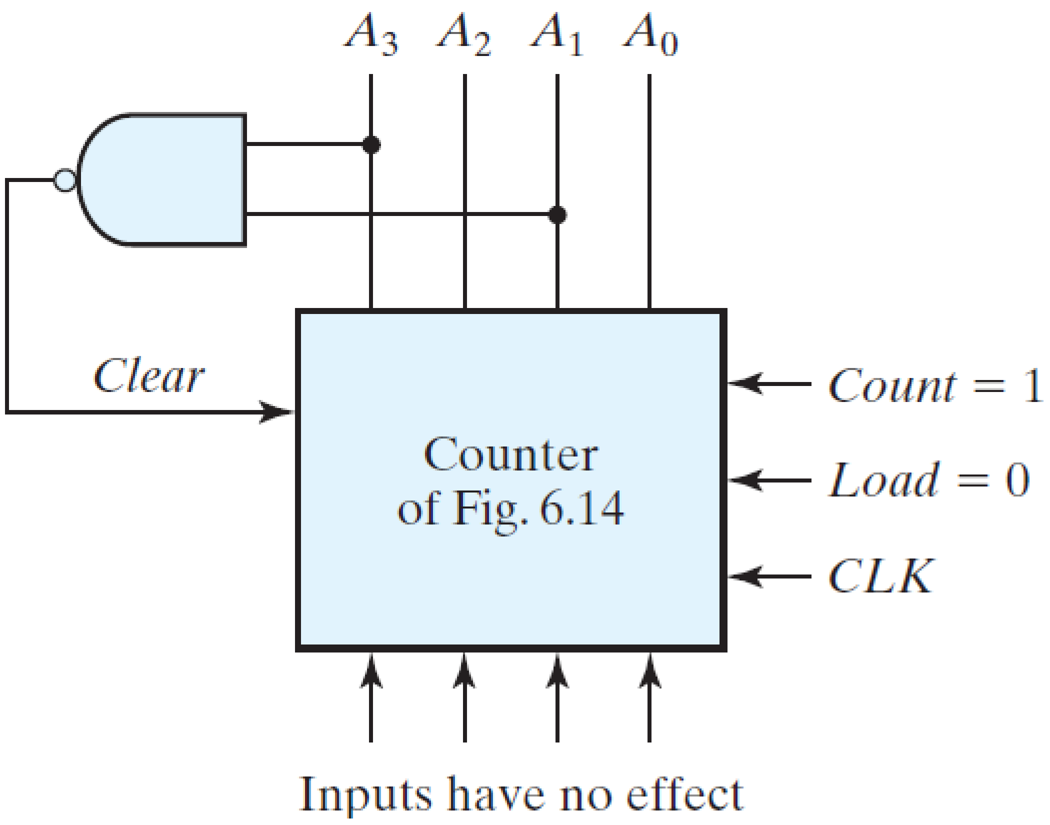


BINARY COUNTER with PARALLEL LOAD

Clear	CLK	Load	Count	Function
0	X	X	X	**CLEAR** to 0
1	$\uparrow$	1	X	**LOAD** inputs
1	$\uparrow$	0	1	**COUNT** to Next Binary state
1	$\uparrow$	0	0	**NO CHANGE**

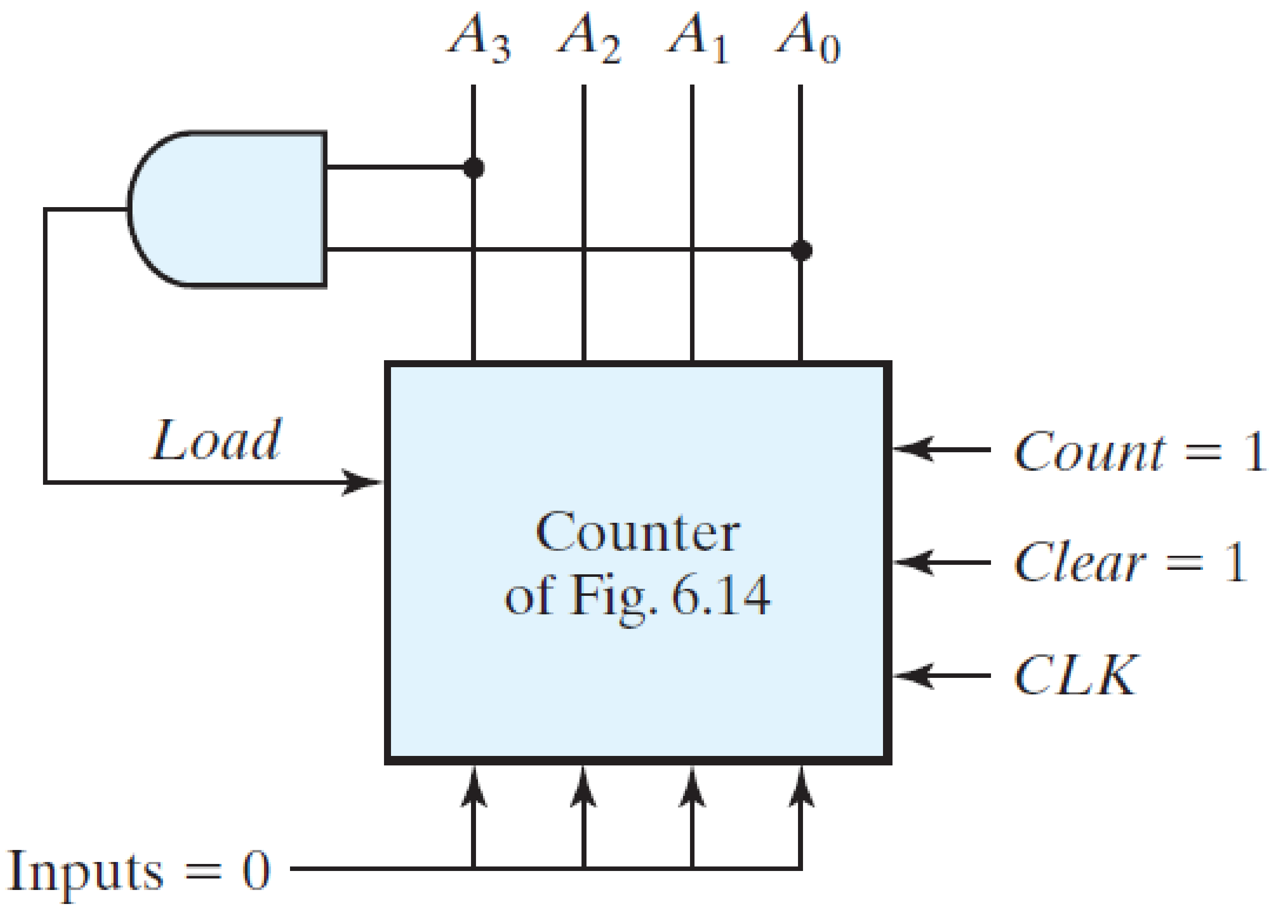
BINARY COUNTER with PARALLEL LOAD as

DECADE COUNTER



BINARY COUNTER with PARALLEL LOAD as

DECADE COUNTER



Thanks